

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2017-2018

CE1005/CZ1005 – DIGITAL LOGIC

Nov/Dec 2017

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. All questions carry equal marks.

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1. (a) (i) Convert the decimal value 1835 to hexadecimal. Show each step clearly.
(2 marks)
 - (ii) Determine the number of bits required to represent an 8-digit telephone number in each case:
 - ASCII code
 - BCD (binary-coded decimal)(2 marks)
 - (iii) A circuit transmits the hexadecimal value ACE6 (which includes one parity bit) with no error. Does the circuit use odd or even parity? State how you derive the answer.
(2 marks)
 - (iv) Determine the signed decimal value of the 8-bit 2's complement number 10010111. Show each step clearly.
(2 marks)

Note: Question No. 1 continues on Page 2

- (b) Simplify the following Boolean expression using algebraic manipulations. Show each step clearly and give your answer as a sum-of-product expression (SOP).

$$F = (w' + x + y + z') (w'y + xz')' (w'y' + xz)'$$

(7 marks)

- (c) Given the Boolean expression:

$$G(a, b, c, d) = (a \text{ XOR } b) \text{ AND } (c \text{ XNOR } d)$$

Obtain both the canonical sum-of-minterm and the canonical product-of-maxterm expressions for G. Show each step clearly.

(5 marks)

- (d) Given the Boolean expression:

$$W = (a + c) (b + d)' (e' f)'$$

Without modifying the expression, sketch a circuit diagram with appropriate logic symbols to show clearly how logic low may be obtained at the output W.

(5 marks)

2. (a) A logic circuit has four inputs: A, B, C*, D* and two outputs: F, G*. * denotes active low signals.

The output F is asserted when

- both A and B are asserted, or
- C* is negated and D* is asserted, or
- both A and C* are asserted, and both B and D* are negated

The output G* is asserted when

- both A and D* are asserted, or
- A is negated and B, C* and D* are all asserted

With the use of Karnaugh maps, obtain the minimum-cost SOP expression for F and the minimum-cost POS expression for G*. Show the loops on the Karnaugh maps clearly.

(10 marks)

Note: Question No. 2 continues on Page 3

- (b) (i) With reference to an inverter, briefly explain the term “propagation delay”.
(2 marks)
- (ii) With reference to a four-bit parallel adder constructed from four full adders, briefly explain the term “carry propagation”.
(3 marks)
- (iii) Briefly describe the purpose of “carry look-ahead” with reference to part Q2(b)(ii).
(3 marks)
- (c) Table Q2 shows the static parameters of two logic devices.

Table Q2

Parameter	Device X	Device Y
V _{cc}	5V	5V
V _{IL(max)}	1.5V	1.8V
V _{OL(max)}	0.7V	0.5V
V _{IH(min)}	3.5V	3.6V
V _{OH(min)}	4.2V	4.5V
I _{IL (max)}	1mA	1.5mA
I _{OL (max)}	18mA	20mA
I _{IH (max)}	0.8mA	1mA
I _{OH (max)}	12mA	15mA

- (i) Calculate each of the following for both device X and device Y:
 • DC noise margin
 • DC fan-out
(4 marks)
- (ii) Determine whether device X is able to drive device Y. Show how you derive the answer.
(3 marks)

3. (a) Let $a = 5'b11011$, $b = 3'b101$ and $c = 3'b011$. Determine the value of $x[6:0]$ after executing each of the following Verilog assign statements.

- assign $x = \{a[4:0], c[2], b[0]\};$
- assign $x = a[4:0];$
- assign $x = b[1] ? \{a, c[1:0]\} : \{b, a[3:0]\};$
- assign $x = b \& c;$

(6 marks)

- (b) Draw the schematic diagram of an implementation of a Full-Adder circuit, using one 3x8 decoder and two OR gates. The sum-of-minterm expressions for the output Sum and Carry are given by:

$$\begin{aligned}\text{Sum}(x, y, z) &= \sum m(1, 2, 4, 7) \\ \text{Carry}(x, y, z) &= \sum m(3, 5, 6, 7)\end{aligned}$$

(7 marks)

- (c) There are three ants each standing at a different vertex of an equilateral triangle (i.e., the triangle's edges have equal lengths). The ants are free to move along the edges of the triangle. You are required to implement a combinational Verilog module which will detect the likelihood of collision of ants. Assume that the module has three 1-bit inputs and one 1-bit output. Each input represents the direction of movement of an ant on an edge of the triangle. The output goes high whenever there is a likelihood of collision. Briefly explain how you arrive at the solution.

(12 marks)

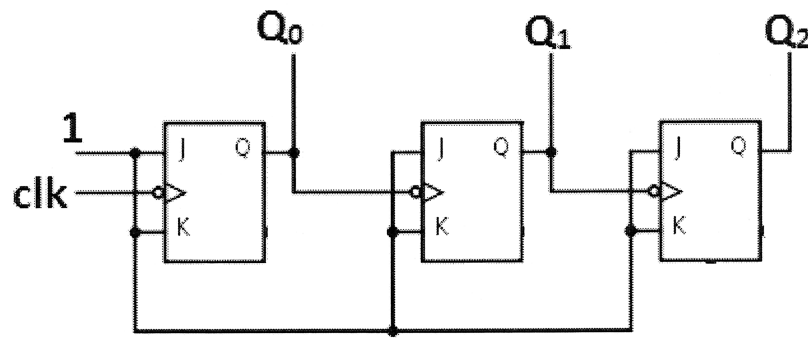
4. (a) What is a SR latch? Briefly describe how it works.

(6 marks)

- (b) Figure Q4a shows a binary ripple counter. Sketch the timing diagram of each of the outputs Q_0 , Q_1 and Q_2 for six clock cycles. Assume the initial state of $Q_2 Q_1 Q_0 = 000$.

(7 marks)

Note: Question No. 4 continues on Page 5

**Figure Q4a**

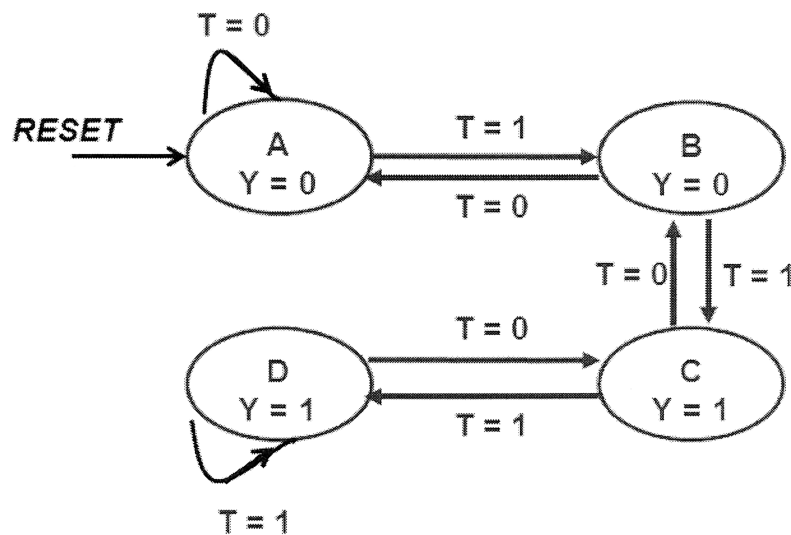
- (c) The state transition diagram of a Finite State Machine (FSM) is shown in Figure Q4b. There are three inputs RESET, CLK and T, and one output Y.

- (i) Determine if it is a Mealy or Moore FSM. Justify your answer.

(2 marks)

- (ii) Write the Verilog module for the FSM shown in Figure Q4b, using an asynchronous RESET.

(10 marks)

**Figure Q4b**

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.